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Decoupling of water and ion dynamics in nanophase-segrated mixtures of an ionic liquid and water studied by NMR experiments and MD simulations

Elisa Steinrücken*, Sebastian Kloth and Michael Vogel

Institute for Condensed Matter Physics, Technische Universität Darmstadt, 64289 Darmstadt, Germany

E-mail: elisa.steinruecken@pkm.tu-darmstadt.de

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Abstract

We combine ^{1}H, ^{2}H, and ^{17}O nuclear magnetic resonance (NMR) experiments with molecular dynamics (MDs) simulations to perform component-selective studies on mixtures of the ionic liquid [C_4mim][DCA] with 50 mol% and 72 mol% water. The computational approach indicates nanophase segregation. While disconnected small water clusters exist in the 50 mol% mixture, a network of water channels forms upon cooling the 72 mol% mixture. The nanophase segregation is accompanied by a rich dynamical behavior. MD simulations, NMR relaxometry, and NMR diffusometry consistently show speedups of reorientation and diffusion at higher water concentration, whereat water motion is faster than cation motion for a given composition. For both components and both mixtures, rotational correlation times and self-diffusion coefficients show a Vogel–Fulcher–Tammann temperature dependence. However, water and cation dynamics decouple upon cooling. Because the degree of decoupling is similar for both mixtures, we conclude that it does not depend on the existence of a network of water channels, but rather results from the formation of soft confinement. A comparison of reorientation and diffusion reveals that the Stokes–Einstein–Debye relation is valid for the cation, whereas it breaks down for water upon cooling. Possible origins of the breakdown in terms of intrinsic properties of supercooled water are discussed.

Supplementary material for this article is available online

Keywords: ionic liquid, water, molecular dynamics, nuclear magnetic resonance, molecular dynamics simulations

* Author to whom any correspondence should be addressed.

1. Introduction

Ionic liquids (ILs) are salts, which have a low melting point, usually below 100 °C, and intriguing chemical and physical properties [1, 2], offering many advantages over other solvents and opening versatile applications [3–8]. The addition of water may strongly change the physicochemical properties of ILs, e.g. their viscosity [9], conductivity [10], glass transition temperature T_g [11], or crystallization behavior [12]. On the one hand, this can be exploited to tune ILs for specific applications [10]. On the other hand, an unwanted uptake of water by hygroscopic ILs from the environment can have unknown side effects [9].

The hydrophilicity of an IL and, thus, its miscibility with water depend on the chemical structure of the anions and cations, e.g. on the fractions of polar and non-polar structural units [13–15]. A possible effect of the addition of water to ILs is nanophase segregation, which significantly affects the properties [16]. Such changes in structural and dynamical properties can have major implications for many applications, e.g. the solvation of biomolecules in IL-water mixtures, where the hydrogen-bond network of water is involved in protein stabilization [17, 18], or as electrolytes in the development of battery materials like zinc [19] or proton batteries [20]. It was found that the structural heterogeneity strongly depends on the water content. In particular, molecular dynamics (MDs) simulations showed that, at low and high water concentrations, the constituents of the minority component form isolated clusters within the matrix of the majority component, while intricate water networks or bicontinuous phases were observed at intermediate concentrations [21–24].

These simulation results are supported by experimental findings. For example, using small-angle scattering and other experimental methods, nanoscopic water clusters were observed in mixtures with ILs comprising 1-butyl-3-methylimidazole [C_4mim] cations and NO_3 [25–29] or BF_4 [30–33] anions. These so-called water pockets with a proposed size of ∼3 nm, were observed at water concentrations of typically ∼70–80 mol%. However, other experimental studies rejected the concept of isolated water pockets at such water concentrations and proposed a continuous water network through the sample [34]. Either way, water is embedded in an IL matrix forming a soft confinement. Therefore, we expect that, under such circumstances, water dynamics is strongly influenced by the geometrical restriction associated with the nanophase segregation, the interplay between hydrogen bonding with electrostatic interactions, and couplings to the fluctuating degrees of freedom of a soft confinement.

Most previous investigations of the MDs in such IL-water mixtures, in particular, of water dynamics focused on temperatures above 0 °C [32, 34–36]. Only very few studies characterized the dynamical behavior down to the glass transition temperature by means of dielectric spectroscopy [11, 37]. It was observed that the ν process of water, which was reported for various aqueous systems [38–40], separates from the α process above T_g and occurs even below the glass transition temperature.

Pabst *et al* [41] studied mixtures of water with an IL comprising [C_4mim] cations and dicyanamide [DCA] anions in a broad temperature range. The structure was investigated by means of small-angle neutron scattering experiments, Raman spectroscopy, and MD simulations. All experimental and computational results revealed a formation of nanosized water clusters for water concentrations between 60 mol% and 72 mol%. For the 72 mol% mixture, the structural α relaxation was studied in a wide temperature range by combining dielectric spectroscopy, dynamic light scattering, nuclear magnetic resonance (NMR) diffusometry, and differential scanning calorimetry. It was found that the mixture behaves like a common glass former featuring a Vogel–Fulcher–Tammann (VFT) temperature dependence of the α process and showing no signs of crystallization for either of the components down to the glassy arrest. Thus, mixtures of water with [C_4mim][DCA] are well suited to study the dynamics of supercooled water clusters interacting with a soft confining matrix over a broad temperature range.

Here, we extend our experimental and computational studies [41] on mixtures of [C_4mim][DCA] and water to investigate the reorientation and diffusion dynamics of the individual components, in particular, of water in a wide temperature range. For this purpose, we use the fact that NMR experiments and MD simulations enable separate studies of ion and water dynamics. In the case of NMR this is achieved by exploiting the isotope selectivity of the method and using suitable isotope labeling, which makes NMR a valuable method to study the properties of ILs and their mixtures [42]. Moreover, we utilize the fact that not only MD simulations provide access to both rotational and diffusive motions but also a combination of NMR relaxometry and NMR diffusometry.

In detail, we study mixtures of [C_4mim][DCA] with 50 mol% and 72 mol% of water. Unlike for the former mixture, our previous analyses revealed extended water clusters for the mixture with the higher water concentration, while water did not crystallize for either of these compositions [41]. Thus, a comparison of the results for these samples should reveal the effects of nanophase segregation on the dynamical behavior. To determine the temperature-dependent structure and dynamics of these mixtures, we complement MD simulations with ^{1}H, ^{2}H, and ^{17}O NMR experiments for selective studies of cation and water dynamics. In doing so, spin-lattice relaxation (SLR) NMR measurements, including field cycling relaxometry (FCR) studies, are used to determine rotational correlation times τ, while static field gradient (SFG) NMR is employed to measure self-diffusion coefficients D. A comparison of all these results provides insights into dynamical (de-)couplings between the components and into the relations between rotational and diffusive dynamics, providing access to the dynamical signatures of the nanophase segregation on dynamics on various time and length scales.

2. NMR background

2.1. SLR

To study cation and water reorientation, we exploit the fact that SLR times T_1 are related to the spectral density $J_2(\omega)$, which is the Fourier transform of the rotational correlation function of the second Legendre polynomial, $F_2(t)$. In the present cases of ^{1}H, ^{2}H, and ^{17}O NMR the relation is given by [43, 44]:

$$\frac{1}{T_1(\omega)} = C_X \left[J_2(\omega) + 4J_2(2\omega)\right]. \tag{1}$$

Here, the constant C_X specifies the strength of the dominating spin interaction (X) and the Larmor frequency ω is proportional to the strength of the applied B_0 field, explicitly, $\omega = -\gamma B_0$, where γ is the gyromagnetic ratio.

In ^{2}H and ^{17}O NMR the quadrupolar (Q) interaction dominates, which describes the interaction of the electric quadrupole moment eQ of the nucleus with the electric field gradient (EFG) at the nuclear site. Then,

$$C_Q = \frac{3\pi^2}{50}\frac{2I+3}{I^2(2I-1)}\chi_Q^2\left(1+\frac{\eta_Q^2}{3}\right) \tag{2}$$

holds, where I denotes the nuclear spin. Moreover, the principal value eq of the EFG tensor determines the quadrupolar coupling constant (QCC), $\chi_Q = eqeQ/h$, while the asymmetry parameter of this tensor is denoted as η_Q. The EFG tensor is linked to the molecular frame. For water, its principal axis points along the O–^{2}H bond in ^{2}H NMR whereas it is orthogonal to the water bonds in ^{17}O NMR. In our ^{2}H NMR studies on the cation, the principle axis is aligned with the labeled C–^{2}H bond of the imidazole ring (see below). Therefore, the quadrupolar interaction depends on the orientation of these entities in our ^{2}H and ^{17}O NMR approaches and the spectral density $J_2(\omega)$ describes the fluctuations of the quadrupolar interaction as a result of their reorientation.

For the C–^{2}H bond of the cation, the EFG tensor is well-defined with $\chi_Q = 167$ kHz and $\eta_Q = 0$ [45, 46]. In ^{2}H and ^{17}O NMR studies on water, the EFG tensor, in general, depends on the properties of the hydrogen-bond network [47–50]. Hence, its parameters, may change with temperature and upon mixing. Because detailed information is lacking for the studied systems, we compare results from two approaches, which are depicted in the supplementary material. In the first approach, we determine the values of χ_Q and η_Q from the ^{2}H and ^{17}O NMR rigid-lattice spectra of water at low temperatures (see figures S1 and S2) and neglect a possible temperature dependence. Specifically, the line-shape analysis yields $\chi_Q = 215$ kHz and $\eta_Q \approx 0$ for ^{2}H and $\chi_Q = 7.0$ MHz and $\eta_Q = 0.75$ for ^{17}O. In the second approach, we use temperature-dependent χ_Q and η_Q values determined in a previous work on bulk water [51], i.e. we neglect changes in the hydrogen-bond networks due to mixing. In the supplementary material (figure S3), we show that the correlation times obtained from both approaches differ by factors of up to 1.3 and 1.5 for ^{2}H and ^{17}O, respectively. Although these discrepancies are noticeable, they are still weak compared to a variation of the correlation times by several orders of magnitude in the studied temperature range so that both approaches yield a similar temperature dependence. In particular, we ensure in the supplementary material (figure S4) that the uncertainties relating to the quadrupolar coupling parameters do not affect our conclusions relating to a breakdown of the Stokes–Einstein–Debye (SED) relation for water, which will be discussed in section 4.4. Therefore, we restrict ourselves to the results from the first approach in the following text.

In ^{1}H NMR approaches to imidazole-based cations, the homonuclear dipole–dipole interaction dominates [52]. Due to the strong distance dependence of this interaction, the main contribution results from the dipolar coupling of pairs of neighboring ^{1}H nuclei in a cation. Under such circumstances, $J_2(\omega)$ probes the reorientation of their internuclear vector and the coupling constant is given by [53]:

$$C_D = \frac{3}{10}\left(\frac{\mu_0}{4\pi}\right)^2\frac{\gamma^4\hbar^2}{r^6}, \tag{3}$$

where γ denotes the gyromagnetic ratio and r is the internuclear distance. However, due to the multiparticle nature of the dipole–dipole interaction, there are also contributions from proton pairs featuring larger distances and, in general, intramolecular dipolar couplings are complemented by intermolecular ones. This means that, in addition to rotational motion, translational motion may cause fluctuations of the homonuclear dipole–dipole interaction. Nevertheless, the spectral density is dominated by reorientation in the frequency ranges of the present analysis [54]. Moreover, the multiparticle interaction results in an elusive effective coupling constant C_D. Here, we utilize that ^{1}H FCR studies allow for a determination of correlation times without explicit knowledge about the value of the coupling constant.

We use various strategies to obtain rotational correlation times τ. First, we exploit the fact that a $T_1(T)$ minimum is found when the correlation time obeys $\tau = 0.616/\omega$, corresponding to reorientation dynamics in the nanosecond regime for typical Larmor frequencies ω. The determination of correlation times at other temperatures relies on information about the shape of $J_2(\omega)$. For glass-forming liquids, dynamical heterogeneity leads to nonexponential correlation functions and correspondingly non-Lorentzian spectral densities. Usually, the Cole–Davidson (CD) function enables an appropriate description [55, 56]:

$$J_2^{CD}(\omega)\,\frac{\sin\left[\beta_{CD}\arctan\left(\omega\tau_{CD}\right)\right]}{\omega\left[1+\left(\omega\tau_{CD}\right)^2\right]^{\beta_{CD}/2}}. \tag{4}$$

Here, τ_{CD} and β_{CD} are the time constant and the width parameter of the CD function, respectively. From the CD parameters, we calculate mean correlation times [57]

$$\langle\tau\rangle = \tau_{CD}\beta_{CD} \tag{5}$$

or peak correlation times [56]

$$\tau_p = \tau_{CD}\left[\tan\left(\frac{\pi}{2\left(\beta_{CD}+1\right)}\right)\right]^{-1}, \tag{6}$$

where the latter describe the peak position of the corresponding CD susceptibility.

At high temperatures, i.e. if $\omega\tau \ll 1$ is valid for all correlation times of a possible distribution, $J_2^{\mathrm{CD}} = \langle\tau\rangle$ holds so that equation (1) can be rewritten as

$$\frac{1}{T_1} = 5C_{\mathrm{X}}\langle\tau\rangle\,. \tag{7}$$

In the supplementary material (figure S5), we use ^{2}H FCR to confirm that, consistent with CD behavior, T_1 shows no dispersion in the high-temperature limit. Therefore, in our ^{2}H and ^{17}O NMR studies, knowledge about the coupling constant C_{X} allows us to determine mean correlation times from $T_1(T)$ at high temperatures without assumptions regarding the exact shape and width of the spectral density $J_2(\omega)$. At lower temperatures, we exploit the fact that the CD width parameter β_{CD} is available from the minimum value of T_1 [55]. Then, assuming that β_{CD} is independent of temperature and using the thus determined spectral density $J_2^{\mathrm{CD}}(\omega)$ in equation (1), correlation times can be obtained from the T_1 times.

FCR allows us to measure $1/T_1(\omega)$ in a broad frequency range and, hence, to directly map out $J_2(\omega)$ [58, 59]. This possibility renders model assumptions about the shape of the spectral density obsolete and enables a straightforward determination of correlation times. For this purpose, we follow previous approaches and transform the T_1 data into a susceptibility representation to mimic the situation in mechanical or electrical relaxation studies [54, 60, 61]. Explicitly, using the fluctuation–dissipation theorem, $\omega J_2(\omega) = \chi_2''(\omega)$, we define the NMR susceptibility

$$\frac{\omega}{T_1(\omega)} = C_{\mathrm{X}}\left[\chi_2''(\omega) + 2\chi_2''(2\omega)\right] \equiv \chi_{\mathrm{NMR}}''(\omega)\ . \tag{8}$$

Then, the positions ω_{p} of susceptibility maxima yield peak correlation times $\tau_{\mathrm{p}} = 0.616/\omega_{\mathrm{p}}$.

2.2. SFG diffusometry

Our SFG magnet provides a magnetic field with a static gradient g along the z axis, i.e. $B(z) = B_0 + gz$. Thus, the Larmor frequency ω of a nuclear spin is dependent on its z-position in the gradient field, $\omega(z) = -\gamma B(z)$. Hence, translational motion leads to a time-dependence of the Larmor frequency. In order to detect this time-dependent behavior, we employ the stimulated-echo (STE) pulse sequence, $90° - t_{\mathrm{e}} - 90° - t_{\mathrm{m}} - 90° - t_{\mathrm{e}}$. The two evolution times t_{e} serve to observe the respective Larmor frequencies, while the mixing time t_{m} allows for diffusive displacements of the molecules. The self-diffusion coefficient D can be obtained by probing the decay of the STE intensity according to

$$S(t_{\mathrm{m}}, t_{\mathrm{e}}) \propto \exp\left[-(\gamma g t_{\mathrm{e}})^2 D\left(t_{\mathrm{m}} + \frac{2}{3}t_{\mathrm{e}}\right)\right]. \tag{9}$$

The inverse length scale of the experiment is determined by the factor $\gamma g t_{\mathrm{e}}$, which can be considered as a generalized scattering vector q [62]. In the present SFG diffusion studies, we probe diffusion on length scales of the order of $q^{-1} \approx 0.1 - 3\,\mu$m. To determine the self-diffusion coefficients D, we measure STE decays $S(t_{\mathrm{m}})$ for multiple fixed evolution times t_{e} and fit these data with equation (9), using a global diffusion coefficient, supplemented by SLR damping, which is obtained from independent measurements. The analysis shows that the STE decays for various evolution times t_{e} and, hence, for various q are consistently described by the same diffusion coefficient D, showing that the model of free diffusion applies. Thus, the nanophase segregation of the studied mixtures does not result in anomalous diffusion on the micrometer scale of SFG diffusometry.

3. Experimental and simulation details

3.1. Samples

We investigate mixtures of the IL 1-butyl-3-methylimidazolium dicyanamide ([C$_4$mim][DCA]), see figure 1, with 50 mol% and 72 mol% of water. In the following text, we will refer to these mixtures as 50 W/IL and 72 W/IL, respectively. For separate NMR studies of water and cation dynamics, we prepare these mixtures with different isotope labeling and take advantage of the isotope selectivity of this method: (i) mixtures comprising the (protonated) IL and $H_2^{17}O$ and (ii) mixtures of a selectively deuterated IL and ^2H_2O. For the former protonated (i) mixtures, water reorientation is selectively investigated in ^{17}O NMR while water and cation are jointly probed in ^{1}H NMR The latter deuterated (ii) mixtures enable selective studies of cation dynamics in ^{1}H NMR whereas both water and cation contribute to the observations in ^{2}H NMR.

For the selective deuteration of the IL, we exploited the fact that the cation chemically exchanges its hydrogen atom at the C2 position of the imidazole ring, see figure 1, with deuterons of ^2H_2O [41, 46, 61, 63]. High-resolution ^{1}H NMR spectra revealed that deuteration degrees between 60% and 83% were obtained in this way. Mixing this selectively deuterated IL with ^2H_2O allows us to single out cation dynamics in ^{1}H NMR because, despite this incomplete deuteration, $\geqslant 98\%$ of the protons are located at cations in the 50 W/IL and 72 W/IL mixtures.

[C$_4$mim][DCA] was purchased from Iolitec GmbH with a specified purity of >98% and dried in a vacuum oven at a temperature of 80 °C and a pressure of less than 1 mbar for at least 24 h before use. ^2H_2O with a purity of 99.9% was purchased from Sigma Aldrich and $H_2^{17}O$ with an ^{17}O enrichment of 39.3% was obtained from Cortecnet. The mixtures were prepared by weighing and their concentrations are given in mol% with an error not exceeding $\pm 1\%$. All samples were flame sealed in glass tubes immediately after their preparation.

3.2. NMR experimental details

The ^{2}H and ^{17}O SLR experiments were carried out on a home-built spectrometer with ^{2}H and ^{17}O Larmor frequencies of $2\pi \cdot 46.7$ MHz and $2\pi \cdot 41.2$ MHz, respectively, and 90° pulse lengths of ~2.5 μs. Further details of this setup can

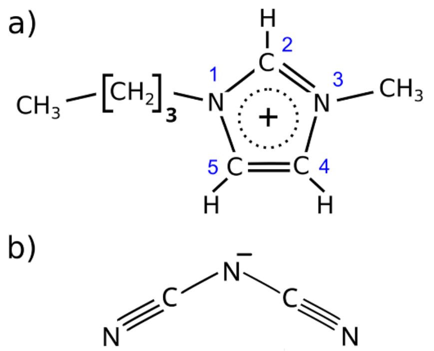

Figure 1. Sketches of (a) the 1-butyl-3-methylimidazole [C_4mim] cation and (b) the dicyanamide [DCA] anion. For the cation the blue numbers label the positions in the imidazole ring. Our deuteration involves the C2 position.

be found in previous works [64–66]. For ^{2}H SLR measurements, we use the saturation–recovery sequence together with a solid-echo detection, while for ^{17}O SLR studies a saturation–recovery sequence was combined with a Hahn-echo detection. In ^{1}H FCR, we utilize a home-built relaxometer, which comprises an electromagnet designed for rapid switching of the magnetic field strength [67, 68]. Larmor frequencies in a range from about $2\pi \cdot 500$ Hz to $2\pi \cdot 30$ MHz were covered. ^{1}H, ^{2}H, and ^{17}O SFG diffusometry were carried out with a customized magnet comprising two superconducting coils in an anti-Helmholtz arrangement to produce a magnetic field with strong gradients [69]. The samples were placed at positions where the ^{1}H, ^{2}H and ^{17}O frequencies amounted to $2\pi \cdot 92.0$ MHz, $2\pi \cdot 23$ MHz and $2\pi \cdot 22$ MHz, respectively, and the field gradient was $g = 179$ T/m for ^{1}H, 148 T/m for ^{2}H and 73 T/m for ^{17}O. For all nuclei, the 90° pulse lengths ranged between 1.0 μs and 3.5 μs in the SFG measurements. In the various setups, the temperature was controlled either by nitrogen gas flow or nitrogen-operated cryostats and stabilized to ± 0.1 K with an absolute accuracy of ± 1 K.

3.3. MD simulation details

The MD simulations were run using GROMACS 2021.6 [70, 71]. The simulated cubic systems comprise 350 ion pairs of the IL and 350 (50 W/IL) or 900 (72 W/IL) water molecules. The force field from Doherty *et al* [72] was utilized for the IL and the TIP4P/2005 model [73] was employed for water. For equilibration, in particular, for density adjustment, the systems were first simulated in the *NpT* ensemble with a pressure of $p = 1$ bar. To ensure appropriate equilibration, the systems were cooled step by step, i.e. the *NpT* simulations at lower temperatures started from an equilibrated structure at the next higher temperature, and the equilibration periods were several times longer than the structural relaxation times, e.g. the systems were equilibrated for 1 μs at the lowest temperatures studied. The production runs were performed in the *NVT* (canonical) ensemble, covering time spans from 100 ns at $T = 300$ K to 2.5 μs at $T = 220$ K.

In all MD simulations, periodic boundary conditions were used and the time step was set to 1 fs. The Coulomb and Lennard–Jones interactions were calculated using the particle-mesh Ewald method [74] with a cutoff radius of 1.3 nm and a Fourier spacing of 0.15 nm. The temperature was controlled with the velocity-rescale thermostat with a time constant of 0.5 ps, while the pressure was controlled with the Parrinello–Rahman barostat and a time constant of 5 ps [75]. Bonds involving hydrogen atoms were constrained with the LINCS algorithm [76] for the cations, while the SETTLE algorithm [77] was used for the water molecules.

4. Results

4.1. Nanophase segregation

First, we employ MD simulations to obtain insights into possible nanophase segregation of [C_4mim][DCA] mixed with 50 mol% and 72 mol% of water, respectively. Characteristic configurations of the simulated mixtures are shown in figure 2. For 50 W/IL, we observe small and isolated water clusters in the IL. For 72 W/IL, the water clusters are larger and more connected in comparison. In particular, a network of water channels forms when the temperature is decreased from 300 K to 220 K. Thus, visual inspection reveals nanophase segregation, which is more prominent for the higher water concentration and lower temperatures. This was confirmed by an analysis of calculated static structure factors in our previous simulation approach [41] and is further corroborated by radial distribution functions in the supplementary material (figures S6 and S7).

For a further quantitative analysis, we determine the probability $P(N_c)$ of finding a water molecule in a cluster comprised of N_c molecules. In doing so, two water molecules are considered to belong to the same cluster if their O–O distance does not exceed 0.35 nm. In figure 3, we see that the water molecules form small clusters with $N_c < 25$ for 50 W/IL at all studied temperatures. By contrast, for 72 W/IL at 300 K, a very broad distribution of cluster sizes is found. Upon cooling, $P(N_c)$ develops a secondary maximum, which sharpens and shifts to larger cluster sizes. At 220 K, ca. 700 of the 900 water molecules belong to the same cluster, indicating that the majority of water molecules form a spanning cluster. This means, on the one hand, that the water molecules form a continuous channel-like network for 72 W/IL, consistent with the results of the above visual inspection, and, on the other hand, that the obtained cluster size looses its meaning because it strongly depends on the system size when a spanning cluster occurs. Altogether, the present analysis confirms our previous experimental and computational findings [41] that 50 W/IL and 72 W/IL differ with respect to nanophase segregation at 300 K and it further shows that the differences are stronger at lower temperatures, where, unlike the former mixture, the latter mixture shows a prominent network of water channels.

4.2. Self diffusion

Next, we study to which degree the different nanostructures of 50 W/IL and 72 W/IL affect the self-diffusion behavior. In MD simulations, we calculate the mean square displacement

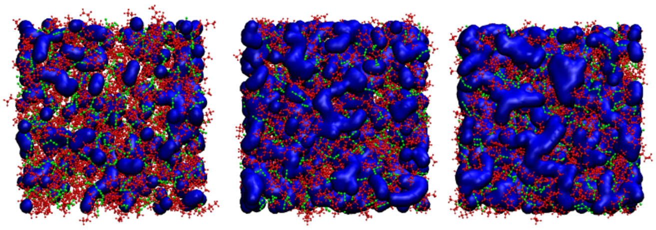

Figure 2. Characteristic configurations from MD simulations: (left) 50 W/IL at 300 K, (middle) 72 W/IL at 300 K, and (right) 72 W/IL at 220 K. The cations are shown in red, the anions in green, and the water molecules are depicted in blue using the QuickSurf algorithm.

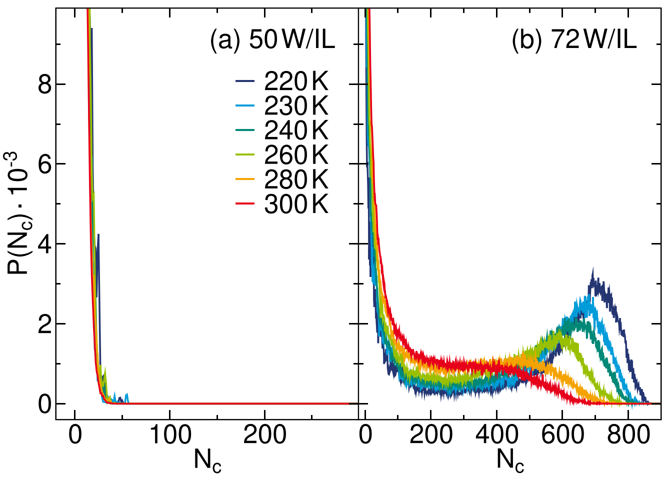

Figure 3. Probability $P(N_c)$ of finding a water molecule in a cluster of size N_c from MD simulations: (a) 50 W/IL and (b) 72 W/IL. For the latter mixture, the data for 300 K was published in previous work [41].

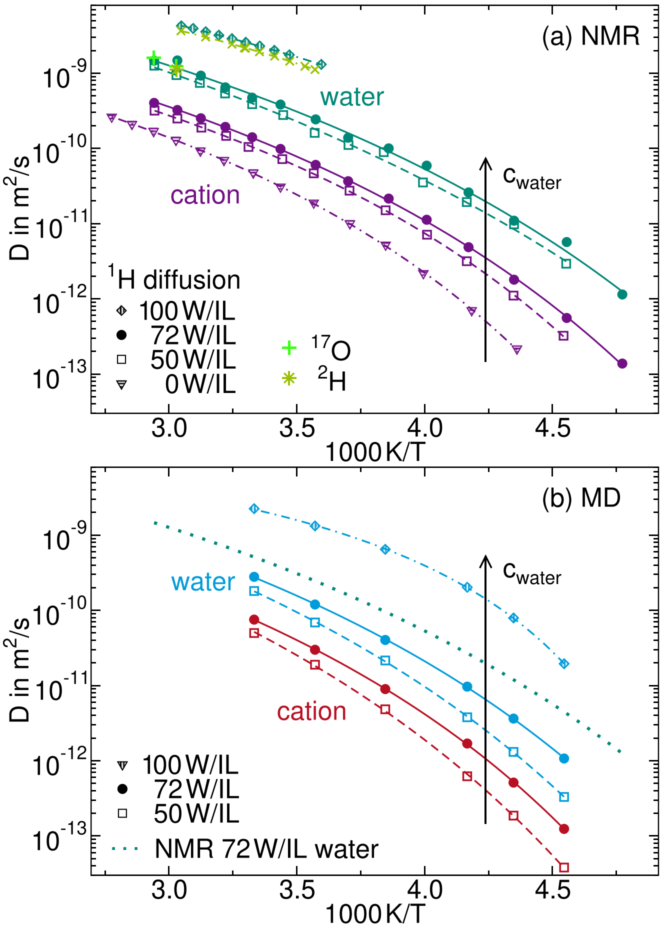

Figure 4. Self-diffusion coefficients D of cation (purple/red) and water (green/blue) in 50 W/IL and 72 W/IL together with the corresponding results for neat [C$_4$mim][DCA] (0 W/IL) and neat water (100 W/IL): (a) D obtained from ^{1}H SFG NMR together with additional ^{17}O (pluses) and ^{2}H (stars) SFG NMR data. For neat water, results for H_2O (diamonds) and D_2O (green crosses) are shown [78]. The cation diffusion coefficients for 72 W/IL were published in previous work [41]. (b) D from MD simulations together with the SFG results for water in 72 W/IL for comparison (dotted line). In both panels, the solid and dashed lines are VFT fits.

$r^2(t)$ of water and cation and determine the self-diffusion coefficients D from the long-time behavior according to $r^2(t \to \infty) = 6Dt$. Exemplary data and fits can be found in the supplementary material (figure S8). In SFG diffusometry, we use suitable probe nuclei and isotope labeling for component selective studies. Exemplary ^{1}H, ^{2}H, and ^{17}O SFG data and fits are also provided in the supplementary material (figure S9). Cation diffusivities are available from ^{1}H SFG studies of mixtures prepared with ^2H_2O, whereas water diffusivities can be obtained from ^{17}O SFG measurements on mixtures prepared with $H_2{}^{17}O$. However, the latter approach is limited to higher temperatures because short ^{17}O T_1 relaxation times hamper a reliable analysis when water diffusion is slower at lower temperatures. Therefore, we extend the temperature range of our studies of water diffusion by applying ^{1}H SFG diffusometry to fully protonated mixtures, where both water and cation contribute to the ^{1}H signal so that a bimodal diffusion decay results. To overcome ambiguities when fitting these data, we use our knowledge from the ^{1}H studies of the deuterated mixtures as external input and fix the diffusion coefficient of the cation at the known value. These fits yield a signal ratio of both components, which is in good agreement with that expected from the proton fractions in the molecular structures (see figure S9). Similarly, we apply ^{2}H SFG diffusometry to the deuterated mixture, for which the ^{2}H signals of both water and cation are detected, and again exploit the knowledge about the cation diffusion coefficient as described above. The diffusion coefficients of water resulting from the different methods are in good agreement, indicating that isotope effects play a negligible role for the diffusion in these mixtures.

Figure 4 shows the water and cation self-diffusion coefficients D obtained from these MD and NMR studies. We see that the water and cation diffusion are faster in mixtures with higher water content. This statement includes the limiting cases of the pure IL (0 W/IL) and pure water (100 W/IL). At a given composition, the diffusivity of water is higher than that of the cation. The temperature dependence of D can be described by the VFT relation $D(T) = D_0 \exp(-B/[T - T_0])$. The VFT fit parameters are compiled in the supplementary material (table SI). Qualitatively, the computational and experimental results yield

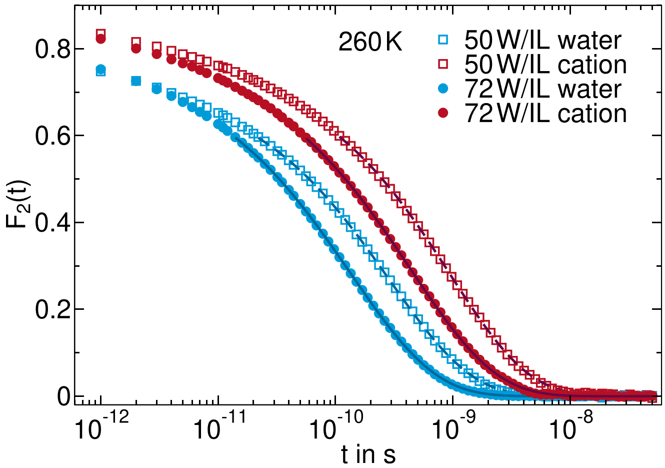

Figure 5. Correlation functions $F_2(t)$ of water molecules (O–H bond) and cation rings (C–H bond at C2 position) calculated from the MD simulations of 50 W/IL and 72 W/IL at 260 K. The lines are stretched exponential fits obtained from the data with $F_2(t) < 0.6$.

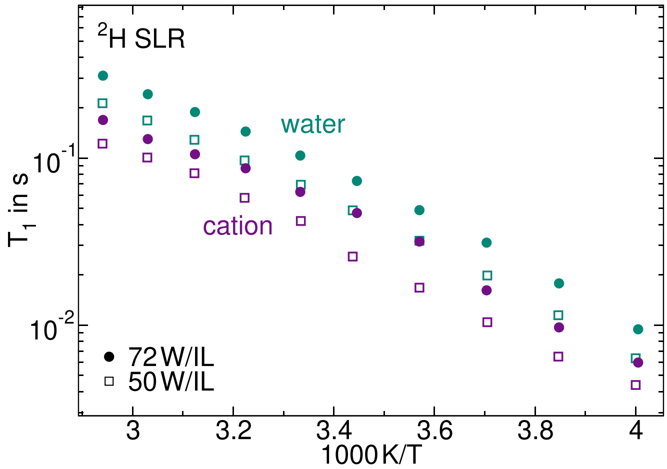

Figure 6. ^{2}H NMR T_1 times of cation and water deuterons in 50 W/IL and 72 W/IL.

a highly consistent picture. Quantitatively, the diffusion coefficients D from MD simulations are a factor of 2–4 smaller than those from NMR measurements, where the difference increases with decreasing temperature.

4.3. Molecular reorientation

Next, we investigate reorientation dynamics based on the rotational correlation functions $F_2(t)$ and the corresponding spectral densities $J_2(\omega)$. In all computational and experimental approaches, we characterize the time scale of the rotational motion by mean correlation times $\langle\tau\rangle$. We compare and discuss all these correlation times together at the end of this section.

In MD simulations, we characterize water and cation reorientation by calculating $F_2(t)$ for the O–H bonds of the water molecules and the C–H bond at the C2 position of the imidazole rings, respectively. Figure 5 shows results for 50 W/IL and 72 W/IL at 260 K. Consistent with the above diffusion results, $F_2(t)$ indicates that water reorientation is faster than cation reorientation and that both speed up with increasing water concentration. For both components and both mixtures and at all studied temperatures, the correlation functions can be described by stretched exponential (Kohlrausch) functions, $F_2(t) \propto \exp[-(t/\tau_K)^{\beta_K}]$, with stretching parameters of $\beta_K = 0.5$–0.6. Thus, the stretching parameters do not yield evidence that different nanostructures of the mixtures lead to different heterogeneity of the reorientation dynamics. From the stretched exponential fits, we calculate mean correlation times by employing the Γ function, explicitly, $\langle\tau\rangle = (\tau_K/\beta_K)\Gamma(1/\beta_K)$.

In NMR experiments, we exploit the relation between the relaxation times T_1 and the spectral density $J_2(\omega)$. In ^{2}H SLR studies, the resonances of the deuterons in C–^{2}H (cation) and O–^{2}H (water) bonds are spectrally resolved in the motional narrowing limit down to 250 K due to their different isotropic chemical shift (see figure S1). This finding means that the deuteron exchange between the components is slow on the experimental time scale and, hence, the T_1 times of cation and water can be determined separately from each other. In all measurements, the buildup of the ^{2}H magnetization follows a single exponential function characterized by the T_1 time. Figure 6 displays the ^{2}H T_1 times for 50 W/IL and 72 W/IL. For both components and both mixtures, T_1 decreases upon cooling, indicative of fast reorientation dynamics with $\tau \ll 1/\omega \approx 1$ ns. In this regime, longer T_1 times indicate faster rotational dynamics. Thus, the ^{2}H SLR results confirm that the reorientation of both components speeds up when increasing the water concentration and, for a given composition, is faster for the water than the cation. For quantification, we determine the mean correlation times $\langle\tau\rangle$ from the ^{2}H T_1 times based on equation (7).

To study water reorientation in a broader temperature range, we perform ^{17}O SLR measurements. Figure 7 presents the ^{17}O T_1 times of 50 W/IL and 72 W/IL. We observe a clear T_1 minimum located at a lower temperature for 72 W/IL than for 50 W/IL, again confirming slower water reorientation for lower water concentration. For both samples, the minimum value of ^{17}O T_1 is higher than the value anticipated for a Lorentzian or, equivalently, Debye spectral density, which would be expected for an exponential correlation function and therefore for dynamics without a distribution of correlation times. Therefore, we use the CD spectral density and determine the width parameter β_{CD} from the height of the T_1 minimum [55]. The analysis yields $\beta_{CD} = 0.32$ and $\beta_{CD} = 0.33$ for 50 W/IL and 72 W/IL, respectively. Using the thus determined spectral densities $J_2(\omega)$ in equation (1), we obtain time constants τ_{CD}, which together with the above width parameters β_{CD} yield the mean correlation times $\langle\tau\rangle$ according to equation (5).

Finally, we selectively investigate cation reorientation in 72 W/IL by means of ^{1}H FCR. The resulting ^{1}H $T_1(\omega)$ dispersions provide access to not only the correlation times of cation reorientation but also the shape of the corresponding susceptibility. In figure 8(a), the ^{1}H FCR data of 72 W/IL are displayed in the susceptibility representation $\chi''_{NMR} \equiv \omega/T_1$.

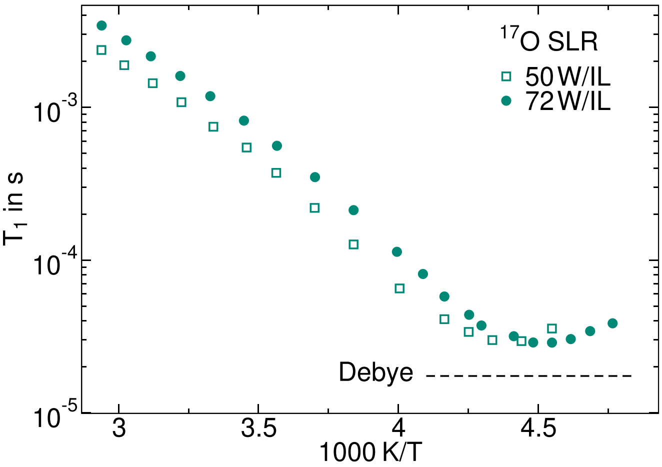

Figure 7. ^{17}O T_1 times of water reorientation in 50 W/IL and 72 W/IL. The black dashed line marks the height of the T_1 minimum calculated for $\omega = 2\pi \cdot 41.2$ MHz, $\chi_Q = 7.0$ MHz and $\eta = 0.75$ assuming a Debye process with a Lorentzian spectral density.

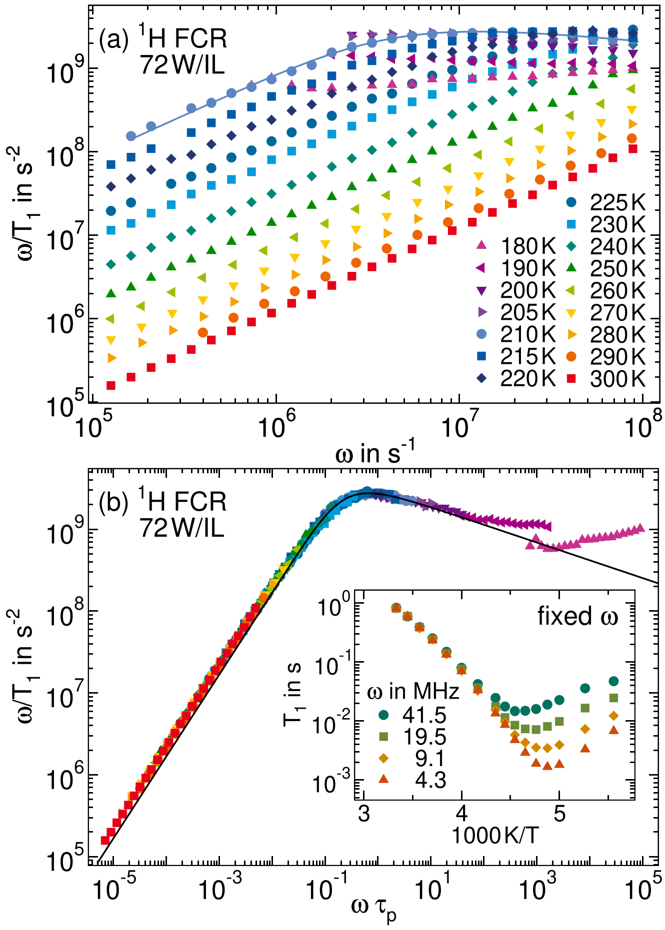

Figure 8. ^{1}H FCR results for cation reorientation in 72 W/IL: (a) NMR susceptibility $\chi''_{\mathrm{NMR}}(\omega) \equiv \omega/T_1$ at the indicated temperatures and (b) susceptibility master curve $\chi''_{\mathrm{NMR}}(\omega\tau_p)$ constructed by shifting the data at various temperatures along the frequency axis. The data below 200 K were rescaled by using extrapolated τ_p values. The inset shows $T_1(T)$ at the indicated fixed Larmor frequencies. In both panels, the solid line is a CD function with a width parameter of $\beta_{\mathrm{CD}} = 0.22$, as obtained from fitting the NMR susceptibility at 210 K.

At high temperatures, we only observe the low-frequency flank of the NMR susceptibility peak in the accessible frequency range, revealing a slope of ω^1, consistent with a CD shape. Upon cooling, the susceptibilities shift to lower frequencies and, eventually, at temperatures below 230 K, the susceptibility peak is located in the experimental window.

For further analysis, we use two approaches. First, we evaluate $T_1(T)$ at several fixed frequencies, as shown in the inset of figure 8(b), and determine peak correlation times τ_p at the respective positions of the T_1 minimum, where $\omega\tau_p = 0.616$ holds. Second, we assume frequency-temperature superposition and obtain peak correlation times from the construction of a master curve. Specifically, we perform a CD fit of the NMR susceptibility at 210 K, which shows a peak well inside the accessible frequency range, to determine the peak correlation time at a reference temperature. Afterward, we shift the NMR susceptibilities at various temperatures along the frequency axis onto the reference curve and calculate temperature-dependent peak correlation times from the respective rescaling factors and the reference value at 210 K.

In figure 8(b), we see that the thus shifted data collapse onto a single susceptibility master curve down to 200 K, while deviations occur below 200 K, most probably, due to the appearance of a previously observed secondary relaxation [61]. Moreover, it is evident that a CD function with a width parameter of $\beta_{\mathrm{CD}} = 0.22$, as obtained from the fit of the susceptibility at 210 K, describes the data in a broad frequency and temperature range, confirming our assumption of frequency-temperature superposition. Some deviations from the CD fit are expected at low frequencies because of additional intermolecular relaxation contributions, which manifest themselves as a shoulder on the low-frequency flank [59]. In the supplementary material (figure S10), we show that the peak correlation times τ_p from both ^{1}H FCR analyses are in good agreement. Instead of these data, we use the mean correlation times $\langle\tau\rangle$ in the following discussion for consistency with the above ^{2}H and ^{17}O results.

In figure 9, we compile the mean correlation times $\langle\tau\rangle$ obtained for water and cation from our experimental and computational approaches. In figure 9(a), we see for both 50 W/IL and 72 W/IL that the ^{17}O and ^{2}H water data agree and that the same is true for the ^{1}H and ^{2}H cation data. These agreements affirm the above analyses and show that isotope effects are negligible for the rotational dynamics. The temperature dependence follows a VFT behavior for both mixtures and components. The resulting fit parameters can be found in the supplementary material (table SII). In both mixtures, water reorientation is about a factor of three faster than cation reorientation, as may be expected from the different molecular sizes and masses. A comparison of the results for 50 W/IL and 72 W/IL reveals that the dynamics of both components is significantly faster for the higher water concentration, again consistent with general expectation and the above qualitative

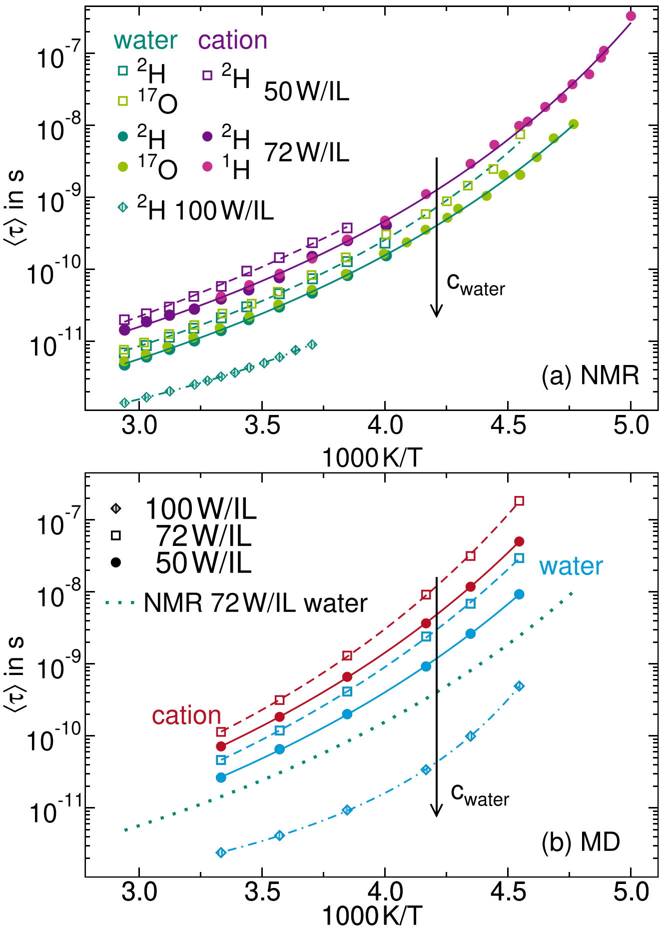

Figure 9. Mean correlation times $\langle\tau\rangle$ of water and cation reorientation in 50 W/IL and 72 W/IL: Results from (a) ^{1}H, ^{2}H, and ^{17}O NMR experiments and (b) MD simulations. In both panels, literature data for bulk water (100 W/IL) are included [79]. The lines are VFT fits. For comparison with the MD data, the VFT fit of the NMR correlation times for water reorientation in 72 W/IL is also included in panel (b) as dotted line.

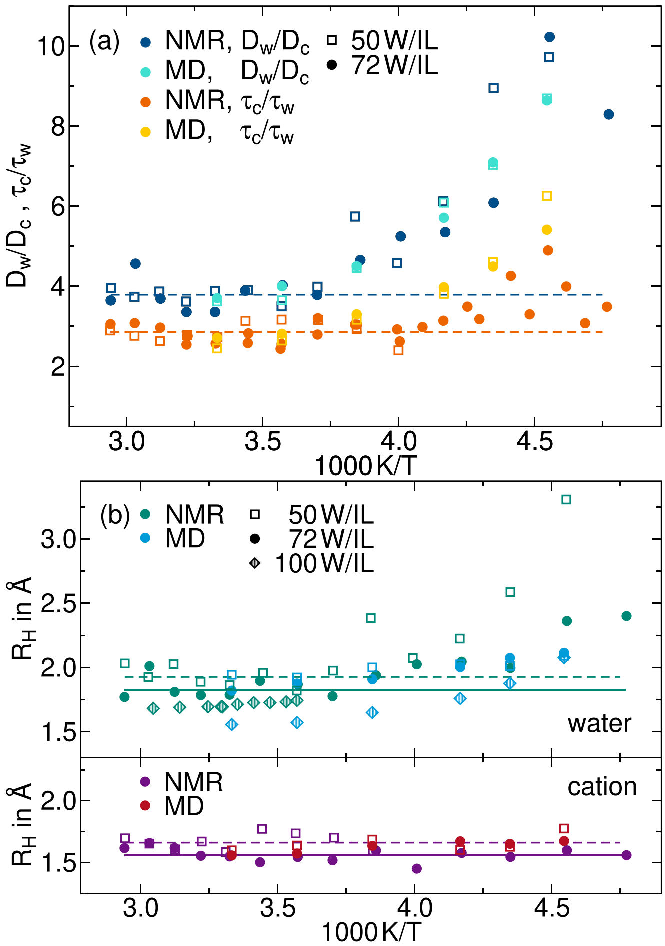

Figure 10. Comparison of various parameters of molecular dynamics in 50 W/IL and 72 W/IL as obtained from the above MD and NMR studies: (a) Ratio of water and cation self-diffusion coefficients, $D_{\mathrm{w}}/D_{\mathrm{c}}$, and ratio of cation and water rotational correlation times $\langle\tau_{\mathrm{c}}\rangle/\langle\tau_{\mathrm{w}}\rangle$. (b) Hydrodynamic radius R_{H} of water and cation calculated from the respective D and $\langle\tau\rangle$ values based on equation (10). For comparison, experimental data calculated from literature data for $\langle\tau\rangle$ [79] and for D [78] as well as MD data for pure water (100 W/IL) are included. In both panels, the lines are guides to the eye.

discussion. In figure 9(b), the MD data show the same trends for the dependencies on the temperature, component, and composition. The VFT fit parameters are also included in table SII. For a quantitative comparison of the experimental and computational correlation times, the NMR data for water reorientation in 72 W/IL are included as the dotted line in figure 9(b). We find that, despite somewhat faster water reorientation in the simulation than in the experiment, the computer model captures the main trends.

4.4. Relations between different dynamical processes

Finally, we investigate temperature-dependent relations between various dynamical processes in the studied mixtures. Specifically, we first compare water and cation dynamics and then relate diffusion and reorientation for each component. In doing so, the dynamic parameters of water and cation are identified by indices 'w' and 'c', respectively.

Figure 10(a) displays the temperature-dependent ratios $D_{\mathrm{w}}/D_{\mathrm{c}}$ and $\langle\tau_{\mathrm{c}}\rangle/\langle\tau_{\mathrm{w}}\rangle$. We observe that our MD and NMR approaches yield highly consistent results, once more confirming that the simulated model well reproduces the experimental reality. All studied ratios hardly depend on the water concentration, i.e. they are not affected by the different nanostructures of 50 W/IL and 72 W/IL. The ratio of the diffusion coefficients amounts to $D_{\mathrm{w}}/D_{\mathrm{c}} \approx 4$ at $T \geqslant 270$ K, while it strongly increases upon cooling, reaching $D_{\mathrm{w}}/D_{\mathrm{c}} \approx 9$ at 220 K. Thus, the diffusivity of water has a much weaker temperature dependence than that of the cations below 270 K. For the ratio of the correlation times, $\langle\tau_{\mathrm{c}}\rangle/\langle\tau_{\mathrm{w}}\rangle$, the effects are qualitatively similar, but less prominent. In particular, water and cation reorientation also decouple at lower temperatures. We note that, in the case of the rotational motion, the experimental findings for the decoupling suffer from some uncertainties arising from limited knowledge about the temperature dependence of the NMR coupling constant, see section 2.1, but the consistency with the simulation data implies that this is a minor effect.

It is tempting to argue that nanophase segregation is at the origin of a decoupling of water and cation dynamics. In particular, one may speculate that fast water diffusion occurs in the network of water channels observed in our simulation approach, see figure 3. Such a scenario was also proposed for water-in-salt electrolytes [80, 81]. However, a similar degree of decoupling is observed in 50 W/IL and 72 W/IL, where a channel-like water network is absent and present, respectively. Hence, although we expect that nanophase segregation is in one way or the other related to the dynamical decoupling, a formation of continuous water channels appears not to be the decisive factor. Moreover, we point out that the decoupling cannot be attributed to the Grotthuss mechanism. Specifically, in ^{1}H NMR diffusometry, Grotthuss-like proton exchange between water molecules may cause enhanced proton diffusion. In the present case, this possibility can be ruled out because a similar decoupling of water and cation diffusivities is observed in MD simulations, where proton exchange does not occur.

The SED relation links molecular diffusion and reorientation in liquids according to:

$$D(T)\,\tau(T) = \frac{2}{9} R_\mathrm{H}^2 . \qquad (10)$$

Here, R_H denotes the hydrodynamic radius of the molecules and the prefactor of $2/9$ considers that F_2 rotational correlation functions are studied in the present MD and NMR approaches. In figure 10(b), we present hydrodynamic radii R_H of cation and water calculated using the SED relation and the above diffusion and reorientation data. Again, the computational and experimental results are consistent, further endorsing both approaches. For the cation, temperature-independent values of $R_\mathrm{H} \approx 1.7$ Å and $R_\mathrm{H} \approx 1.6$ Å are found for 50 W/IL and 72 W/IL, respectively. However, the value of R_H is smaller than expected from the molecular size of the cation. For the water, R_H is approximately constant for $T > 250$ K with values of $\sim$1.9 Å and $\sim$1.8 Å for 50 W/IL and 72 W/IL, respectively, indicating that the SED relation is obeyed. These values are similar to $R_\mathrm{H} \approx 1.6$ Å and $R_\mathrm{H} \approx 1.7$ Å for pure water, which we calculated from available computational and experimental data [78, 79], respectively. However, at lower temperatures, the R_H of water increases in the studied mixtures. For supercooled liquids [82], in particular, for hydrogen-bonded ones [59, 83], SED analysis often yields a hydrodynamic radius, which does not correctly measure the size of the diffusing and rotating molecules. This is also the case here, where the observed increase of R_H implies a mild decoupling of diffusion and reorientation. Explicitly, it indicates that, upon cooling, water diffusion slows down less than water reorientation, a situation occurring for neat glass formers only at lower temperatures [82].

An SED breakdown was also found in previous MD simulation studies of pure water [84–87]. Therefore, one may speculate that it is an inherent property of water. However, different origins of the breakdown were proposed. Some authors attributed the effect to dynamical heterogeneity, explicitly, it was argued that fast and slow water fractions contribute to diffusion and reorientation differently [86]. Others related the SED breakdown to the crossing of the Widom line, which extends the coexistence curve of the proposed high-density liquid (HDL) and low-density liquid (LDL) water phases into the one phase region, or, in other words, to an evolution from HDL-like to LDL-like structural motifs upon cooling [87].

5. Conclusions

We performed MD simulations and NMR experiments to investigate mixtures of [C$_4$mim][DAC] with 50 mol% (50 W/IL) and 72 mol% (72 W/IL) of water. Consistent with previous experimental and computational studies of these mixtures [41], our results indicated a prominent nanophase segregation. Specifically, water was arranged in disconnected small clusters in 50 W/IL, whereas a network of water channels formed upon cooling in 72 W/IL. This nanophase segregation of the mixtures was accompanied by complex MDs.

To explore the dynamical behavior, we exploited the fact that MD simulations and NMR measurements enabled component-selective studies of both reorientation and diffusion. Experimentally, such comprehensive analysis was achieved by performing ^{1}H, ^{2}H, and ^{17}O NMR studies of mixtures with appropriate isotope labeling and by combining NMR relaxometry and NMR diffusometry. The computational and experimental results consistently revealed a speedup of dynamics at higher water concentration, while water motion was faster than cation motion for a given composition. Moreover, for both components and both mixtures, rotational correlation times and self-diffusion showed a VFT temperature dependence. The detailed investigation of the dynamics, in particular the diffusivity, for separate components of the mixtures further yields important information for applications of IL-water mixtures, e.g. in the design of electrolyte batteries, where knowledge of long-range transport properties is essential.

A closer analysis of the relations between various dynamical processes yielded interesting insights. In particular, we found that water and cation dynamics decoupled upon cooling. This decoupling was clearly evident from a significant and continuous increase of the ratio between the water and cation diffusivities below $\sim$250 K. Thereat, the MD and NMR data fully agreed. We propose that the effect is related to the nanophase segregation of the mixtures. Because the degree of decoupling was similar in 50 W/IL and 72 W/IL, we conclude that the formation of a network of water channels is not the decisive factor. Rather, we expect that we observe the onset of a crossover from coupled water and ion motions in the high-temperature mixtures to water dynamics in a less mobile but still soft confinement. Comparing reorientation and diffusion, we found that the SED relation was valid for the cation, whereas it failed for water upon cooling. Because a similar SED breakdown was observed in previous MD simulations of pure water, we assume that it is an intrinsic water property. In particular, it deserves further attention whether or not the effect can be attributed to an evolution from HDL-like to LDL-like water structures.

Altogether, our results suggest that IL-water mixtures are suitable model systems to investigate the properties of supercooled water in soft confinements, where the behavior may more closely resemble that of bulk water in the no man's land than in hard confinements.

Data availability statement

The data that support the findings of this study are openly available at the following URL/DOI: https://doi.org/10.48328/tudatalib-1801 [88].

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Supplementary material

See the supplementary material for ^{2}H and ^{17}O NMR spectra of the studies mixtures, an alternative approach for obtaining ^{2}H and ^{17}O correlation times from SLR analysis with a temperature dependent QCC as well as the SED relation calculated with these correlation times, pair distribution functions and mean square displacements of water and cation in both IL-water mixtures from MD simulations, exemplary SFG STE decays of the 72 mol% mixture measured for ^{1}H, ^{2}H and ^{17}O nuclei at 270 K, 330 K and 340 K, respectively, ^{2}H frequency dependent ^{2}H T_1 of deuterated 72 W/IL and the VFT fit parameters obtained from temperature-dependent analysis of diffusion and reorientation.

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